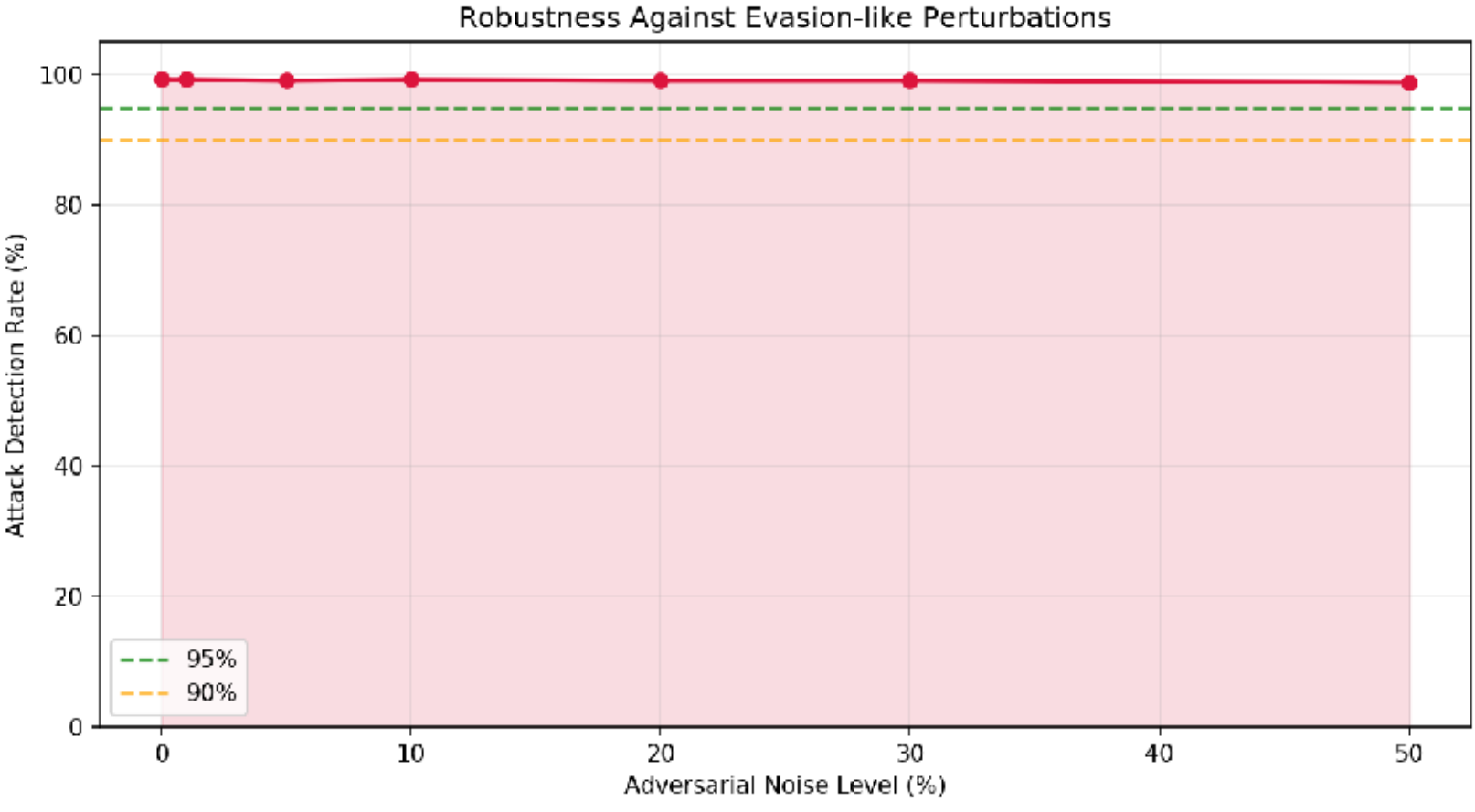
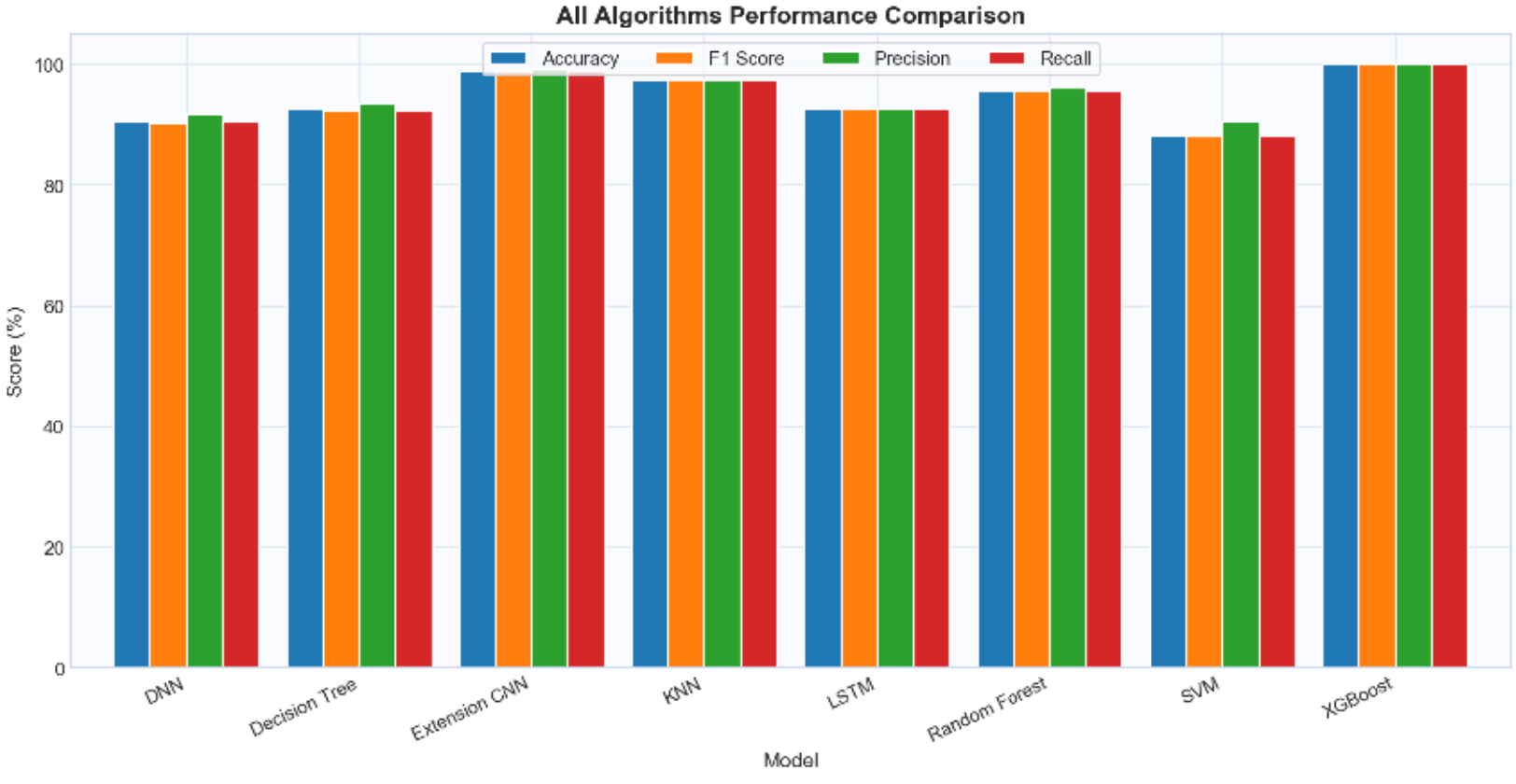
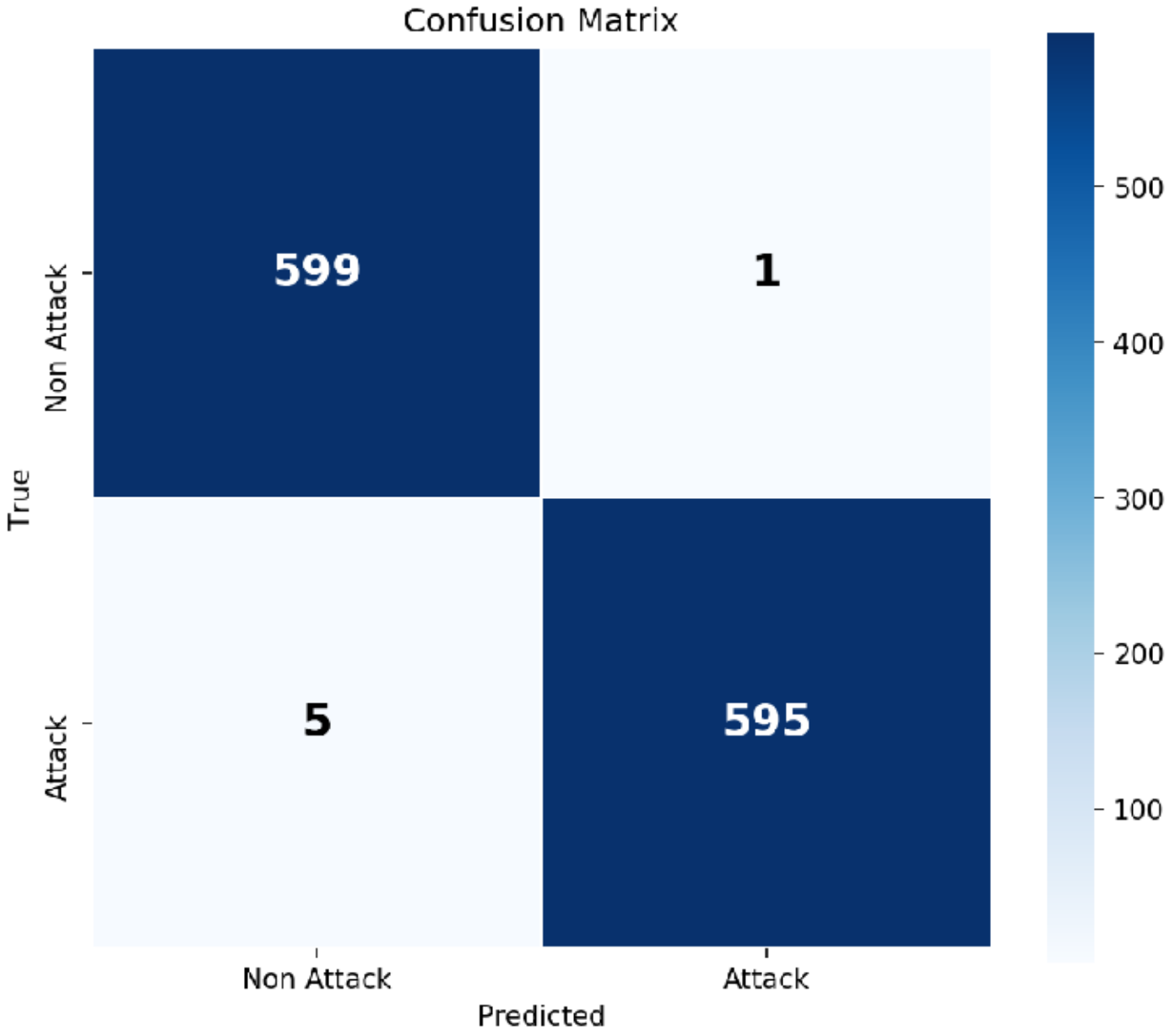
adversarial_robustness.png



all_algorithms_performance_graph.png

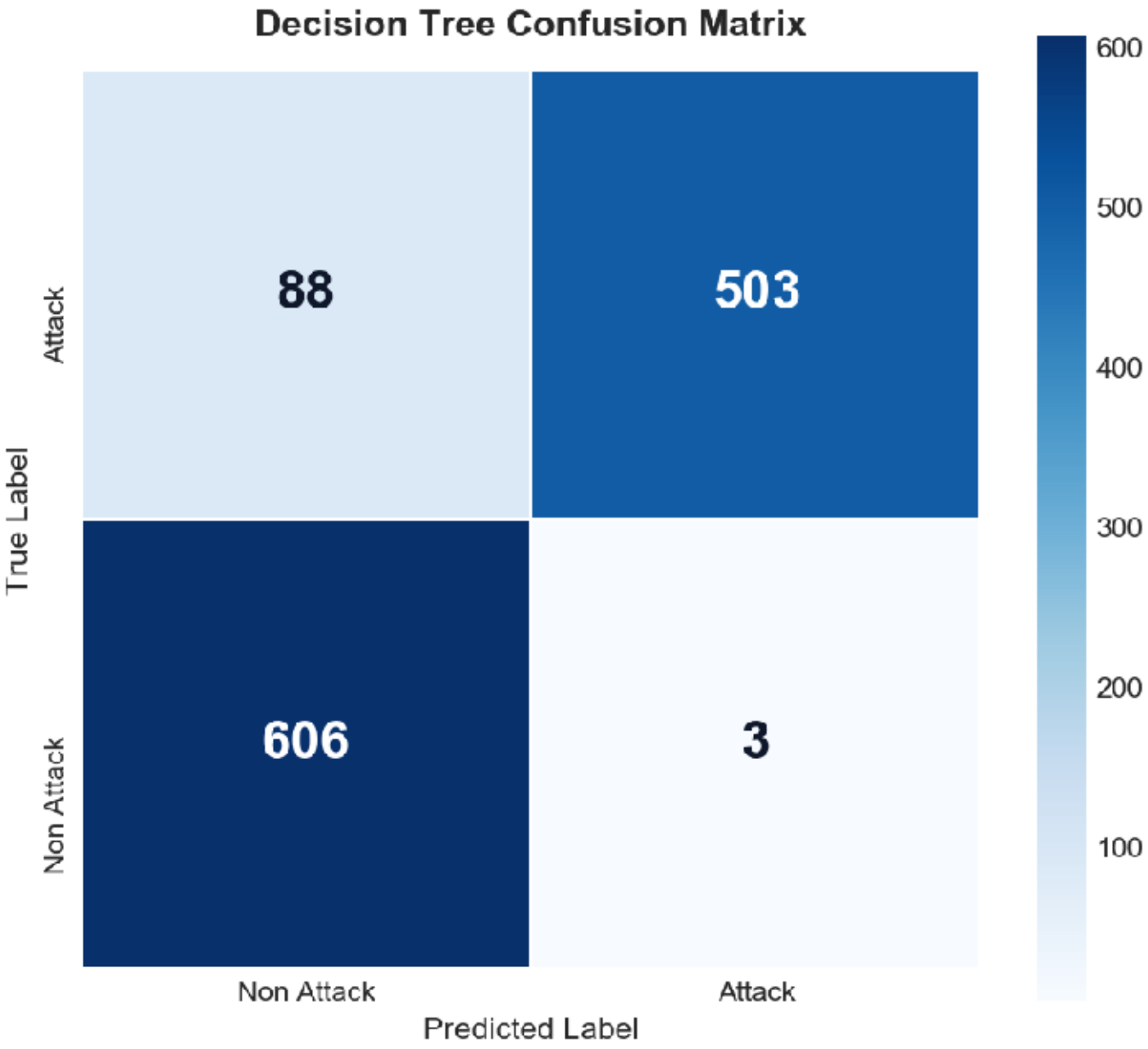


baseline_confusion_matrix.png

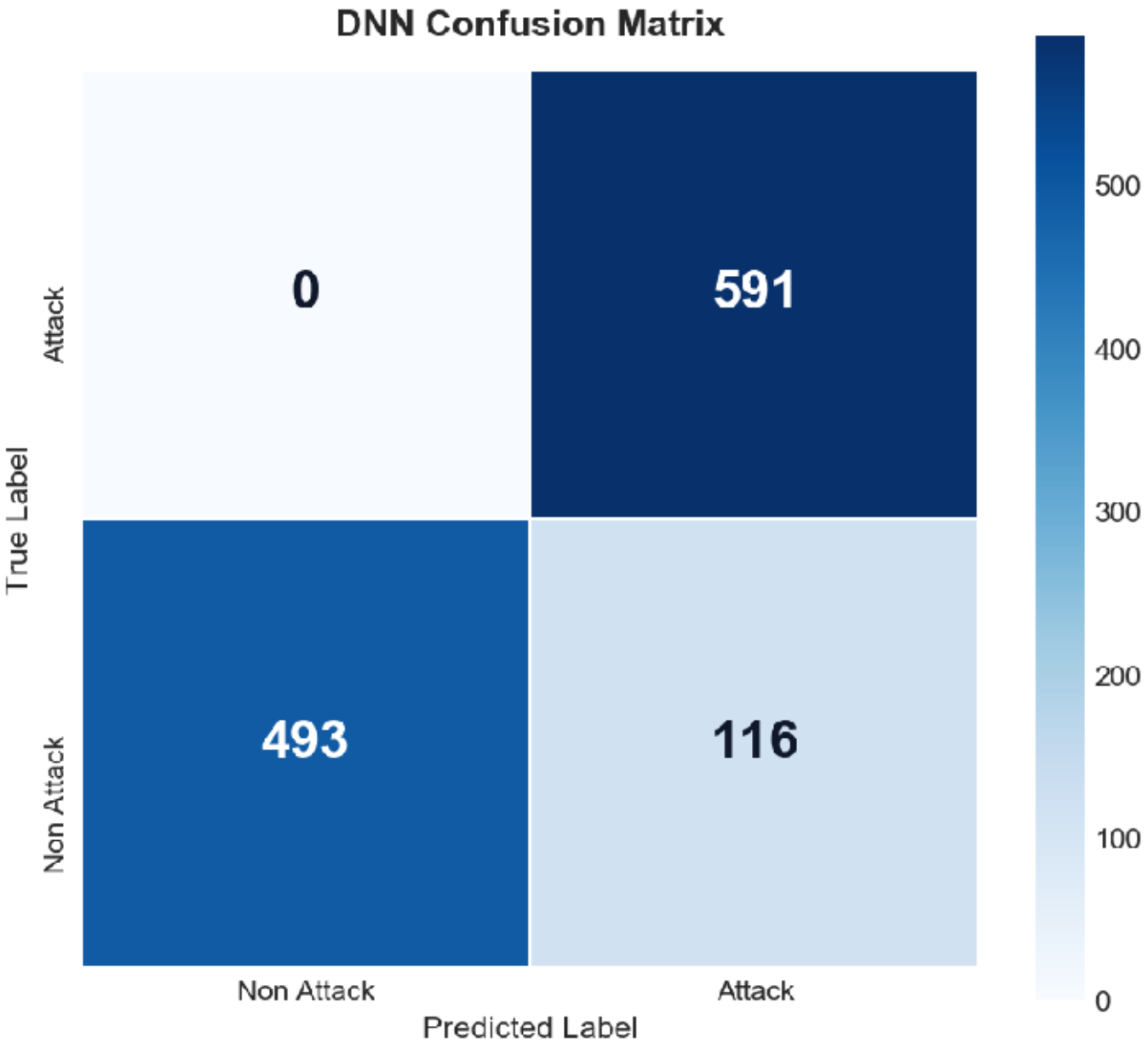


dataset_class_distribution.png

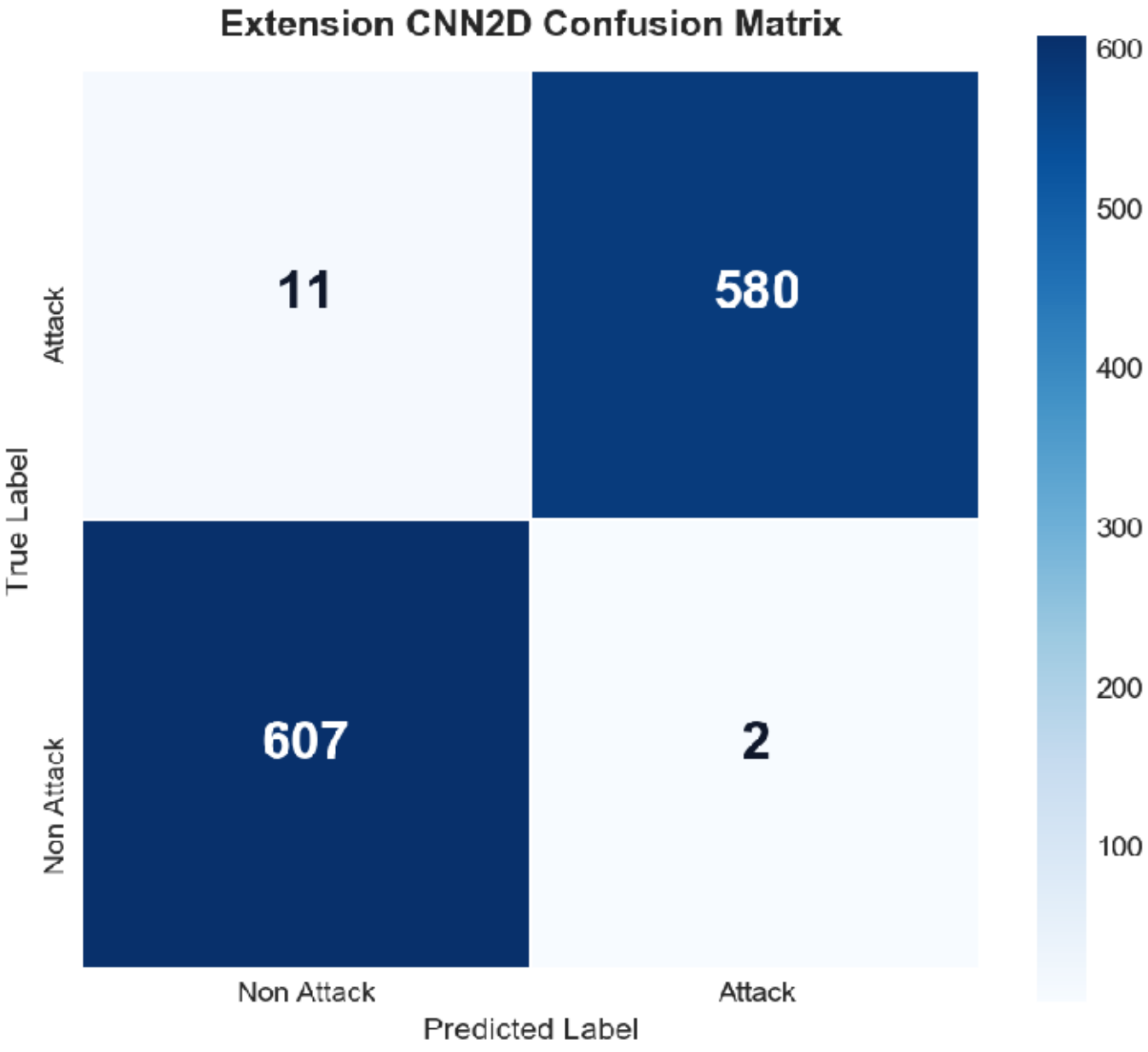
decision_tree_confusion_matrix.png



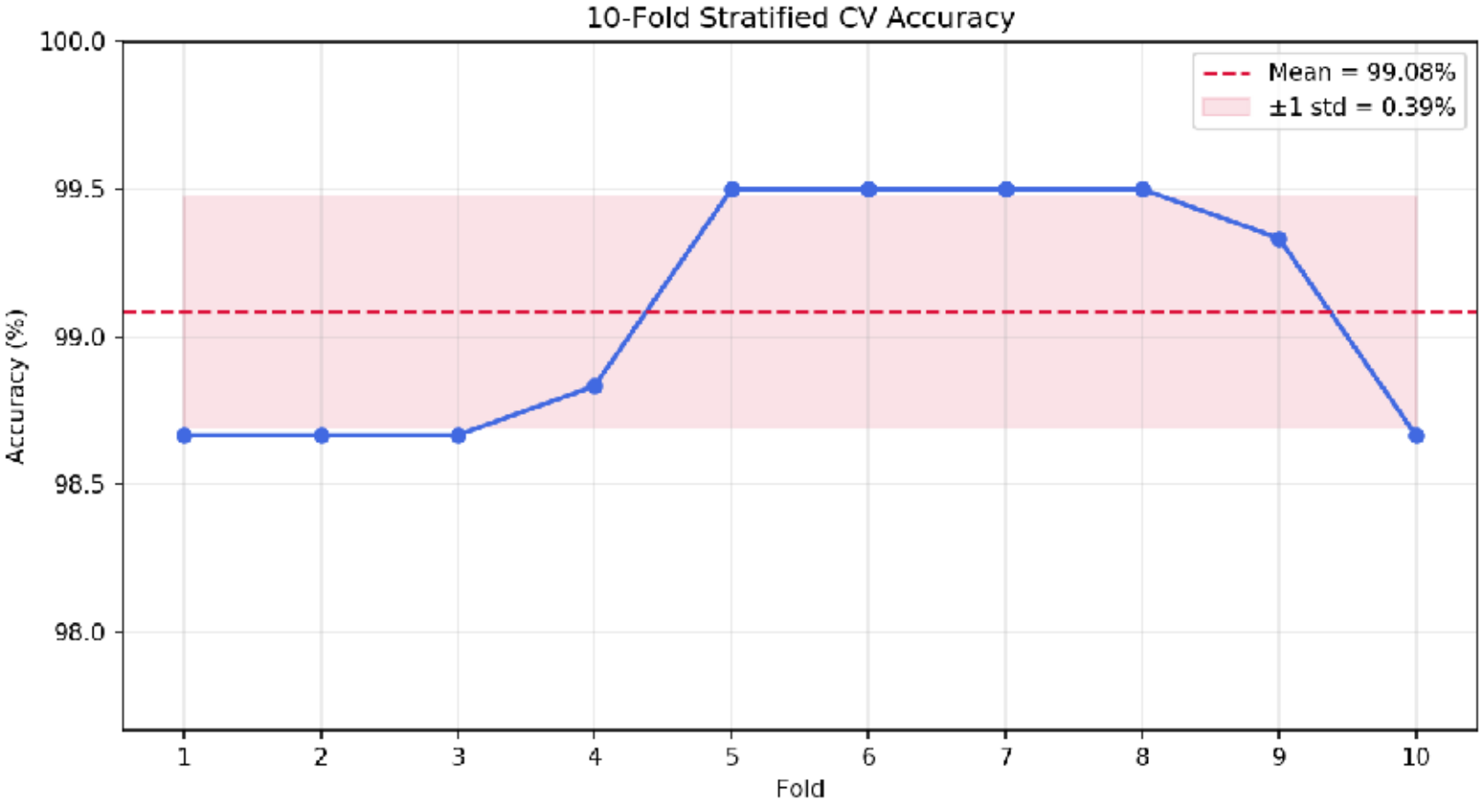
dnn_confusion_matrix.png



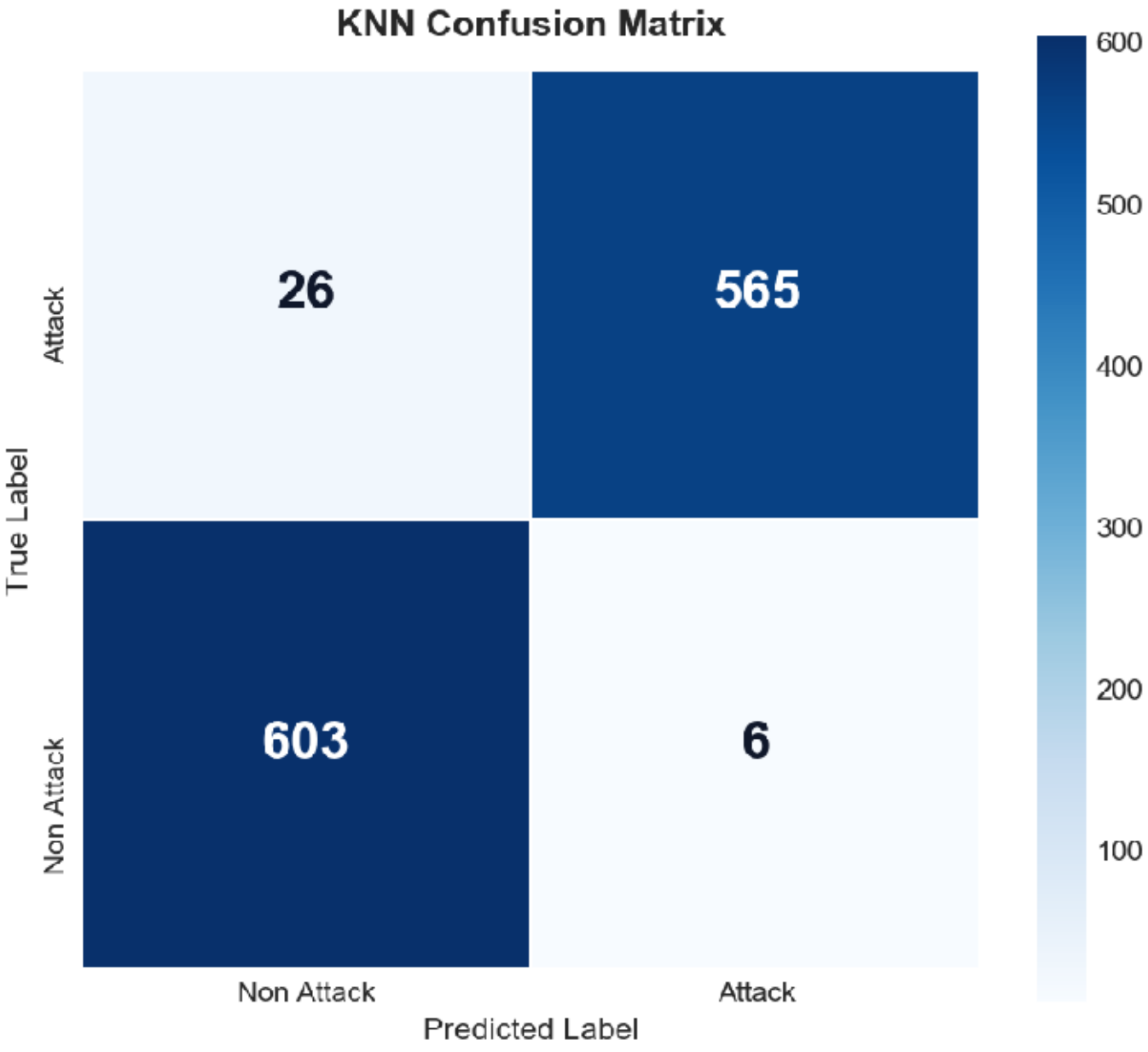
extension_cnn2d_confusion_matrix.png



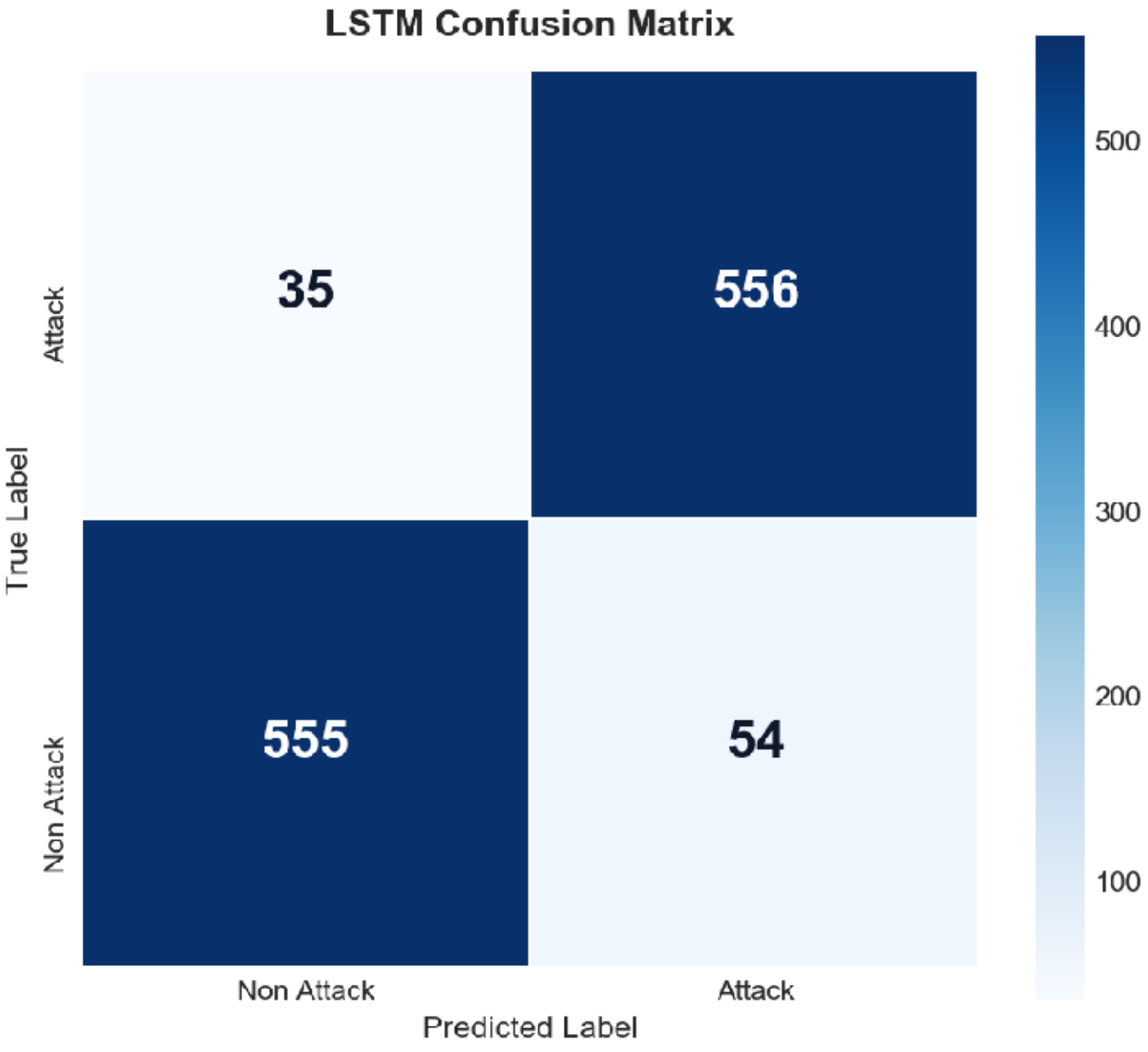
kfold_cv_accuracy.png



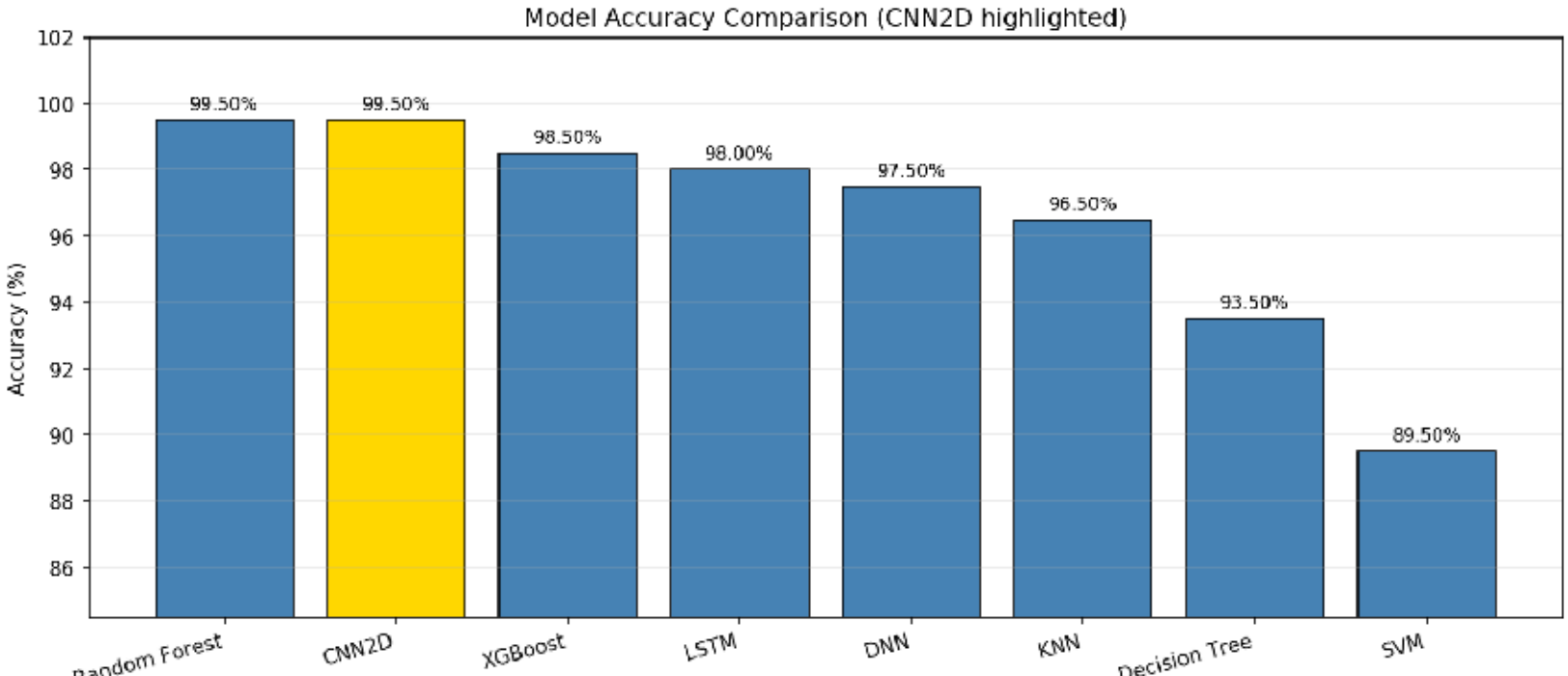
knn_confusion_matrix.png



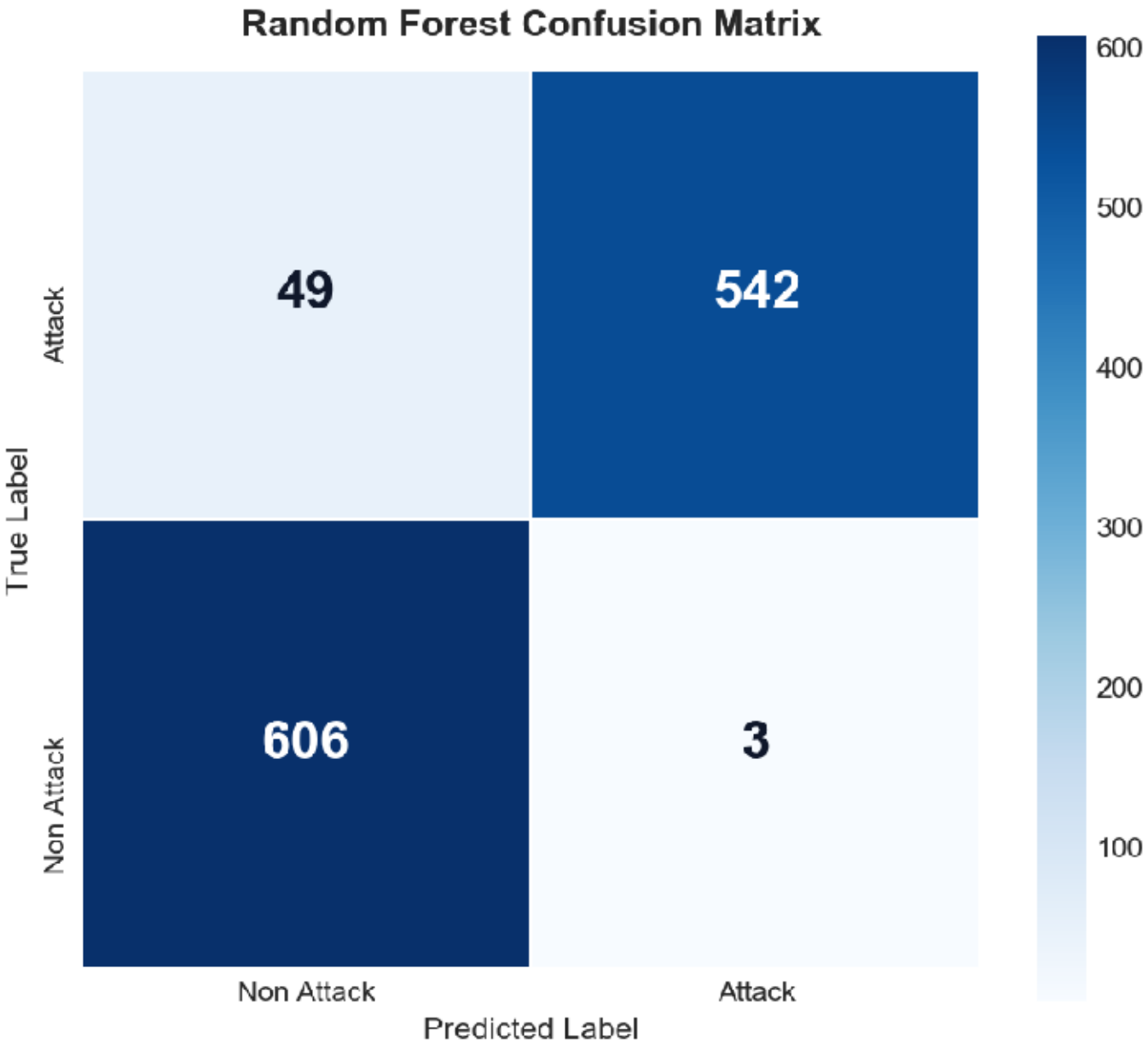
lstm_confusion_matrix.png



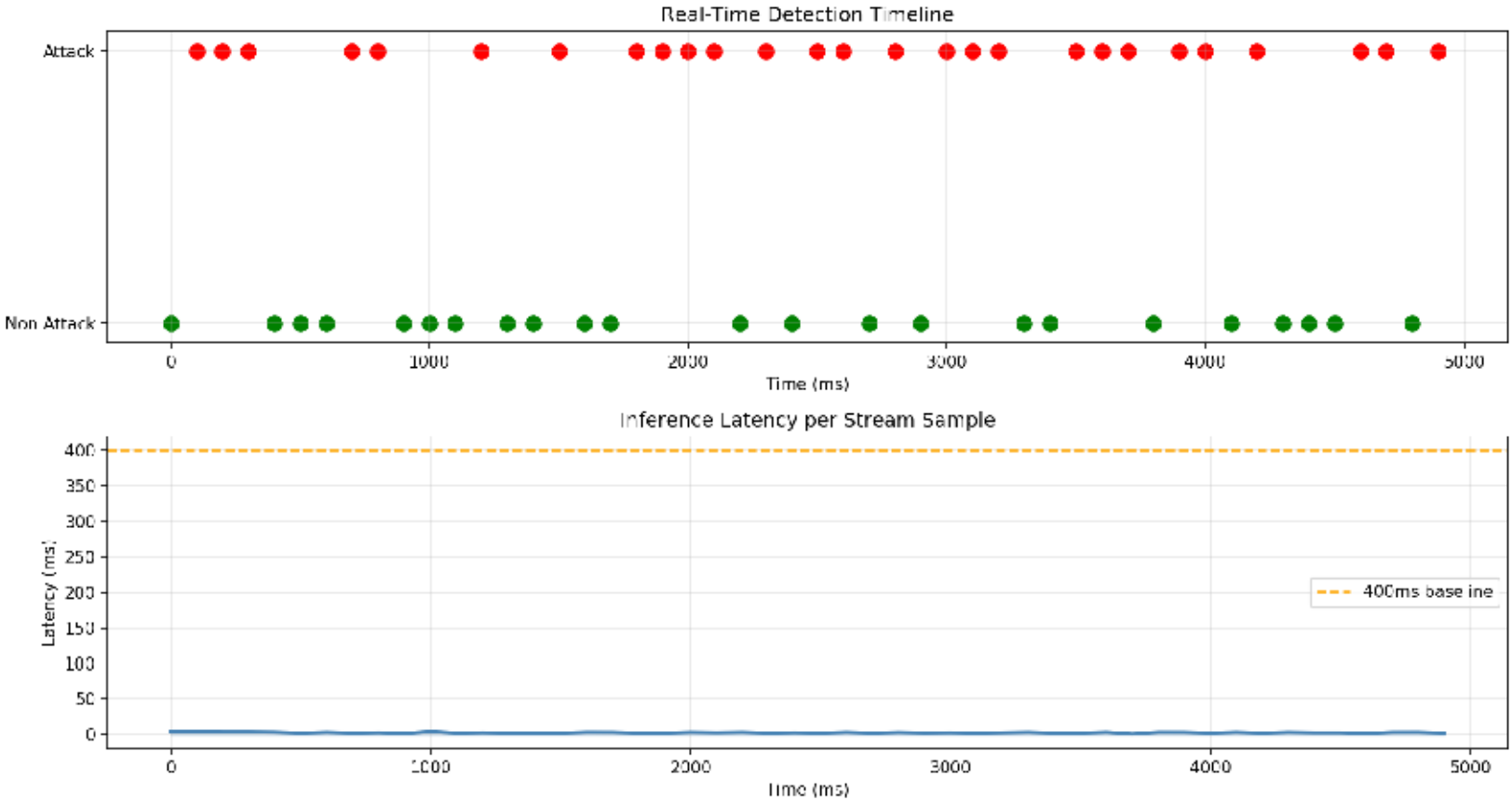
model_comparison.png



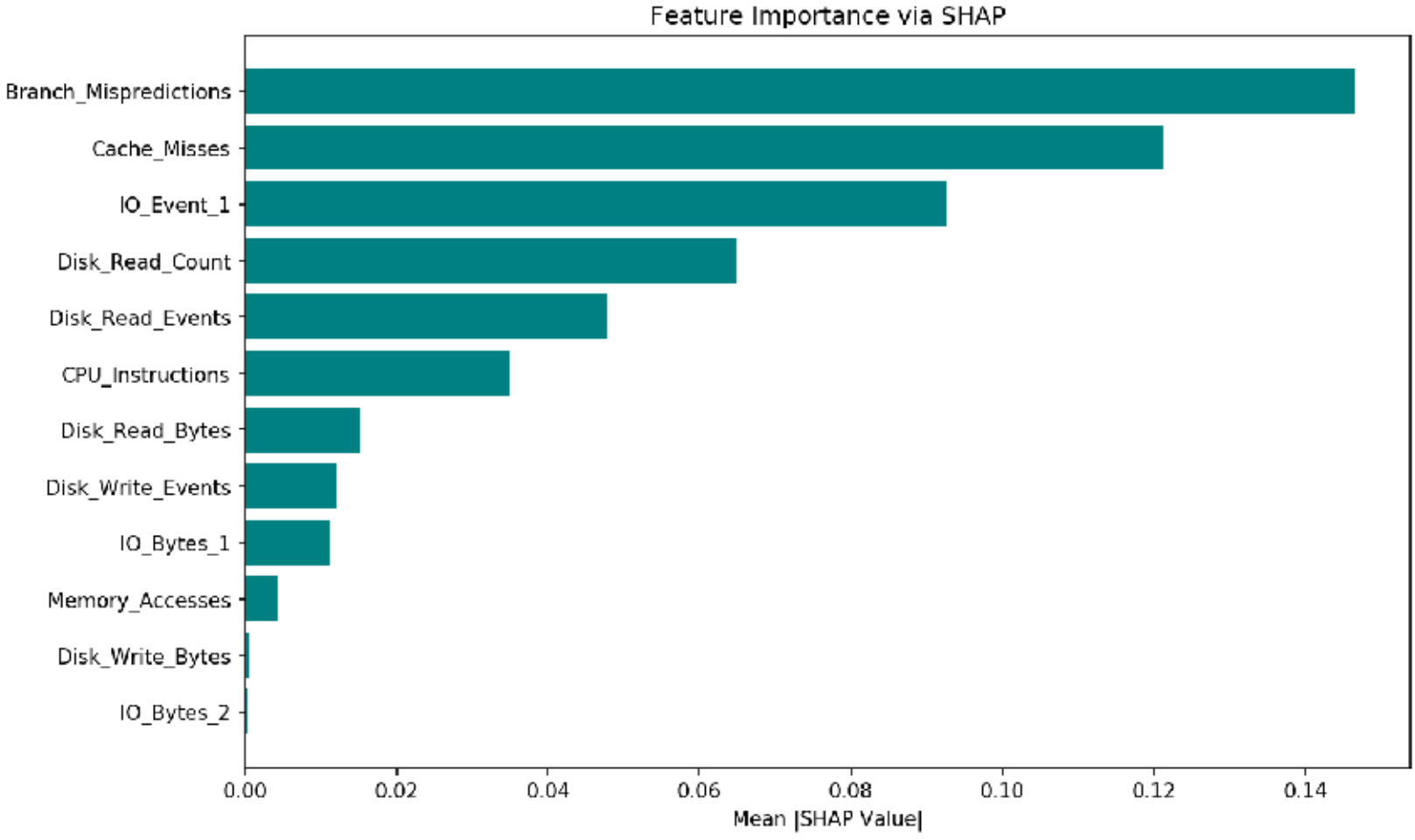
random_forest_confusion_matrix.png



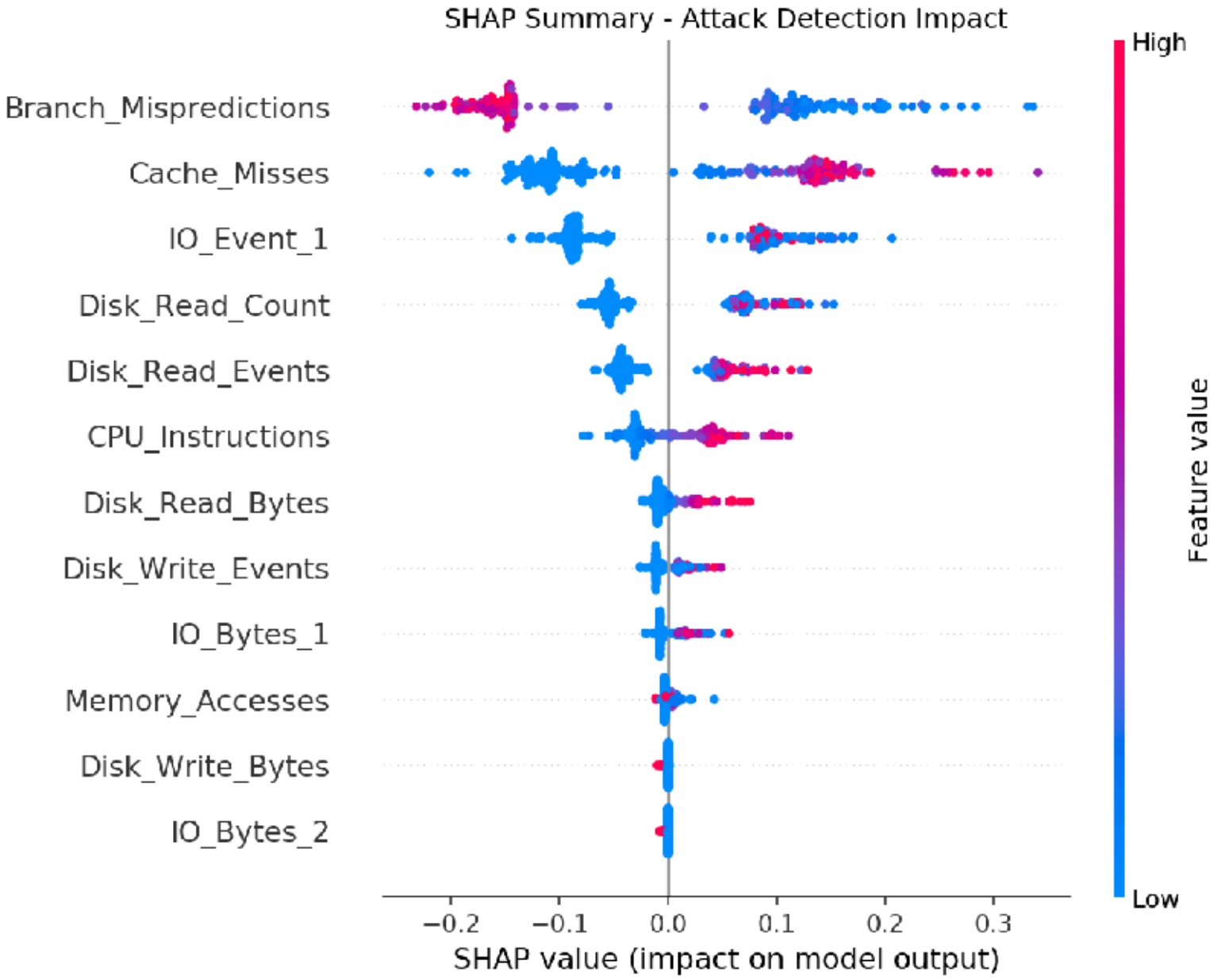
realtime_simulation.png



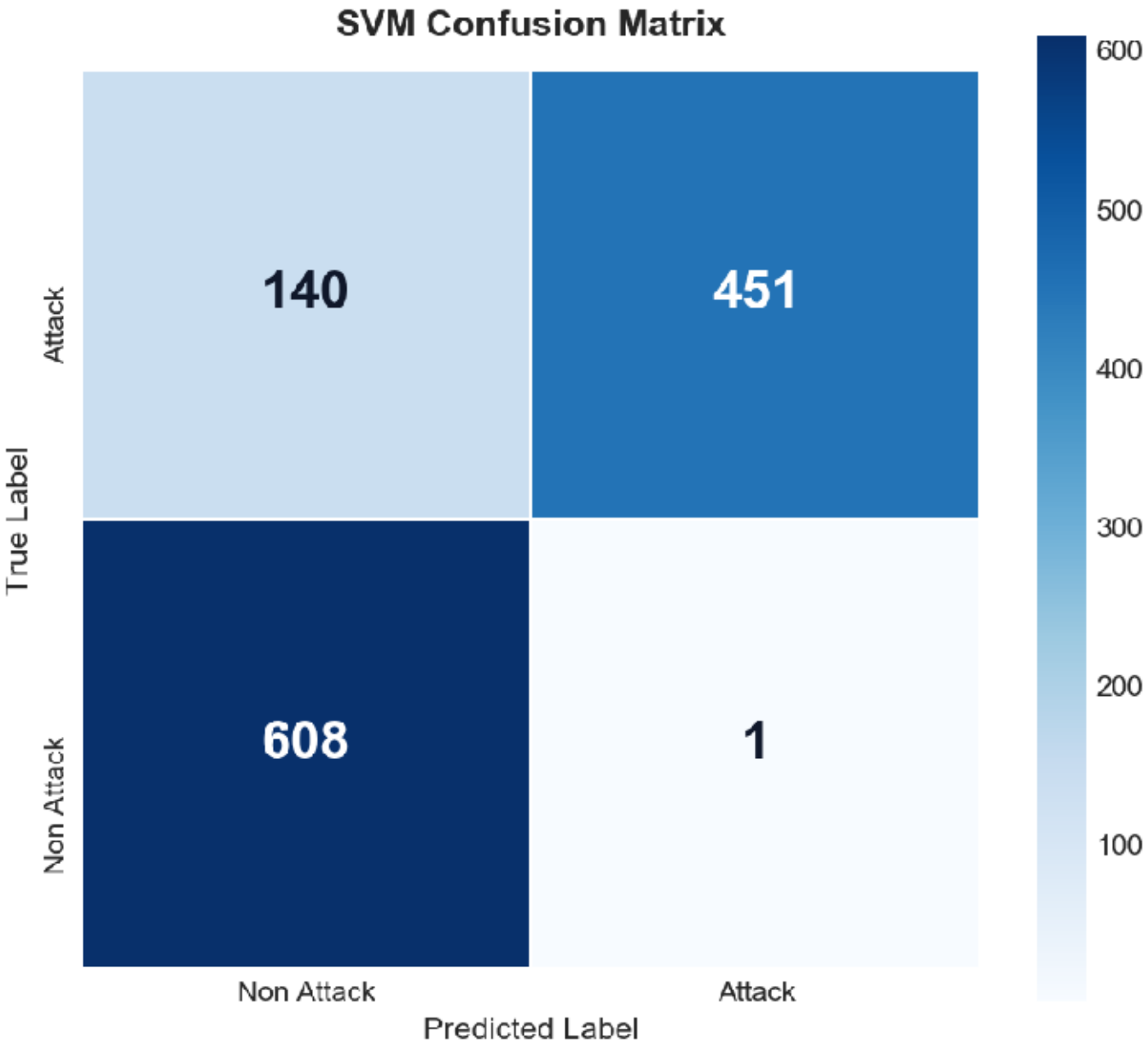
shap_feature_importance.png



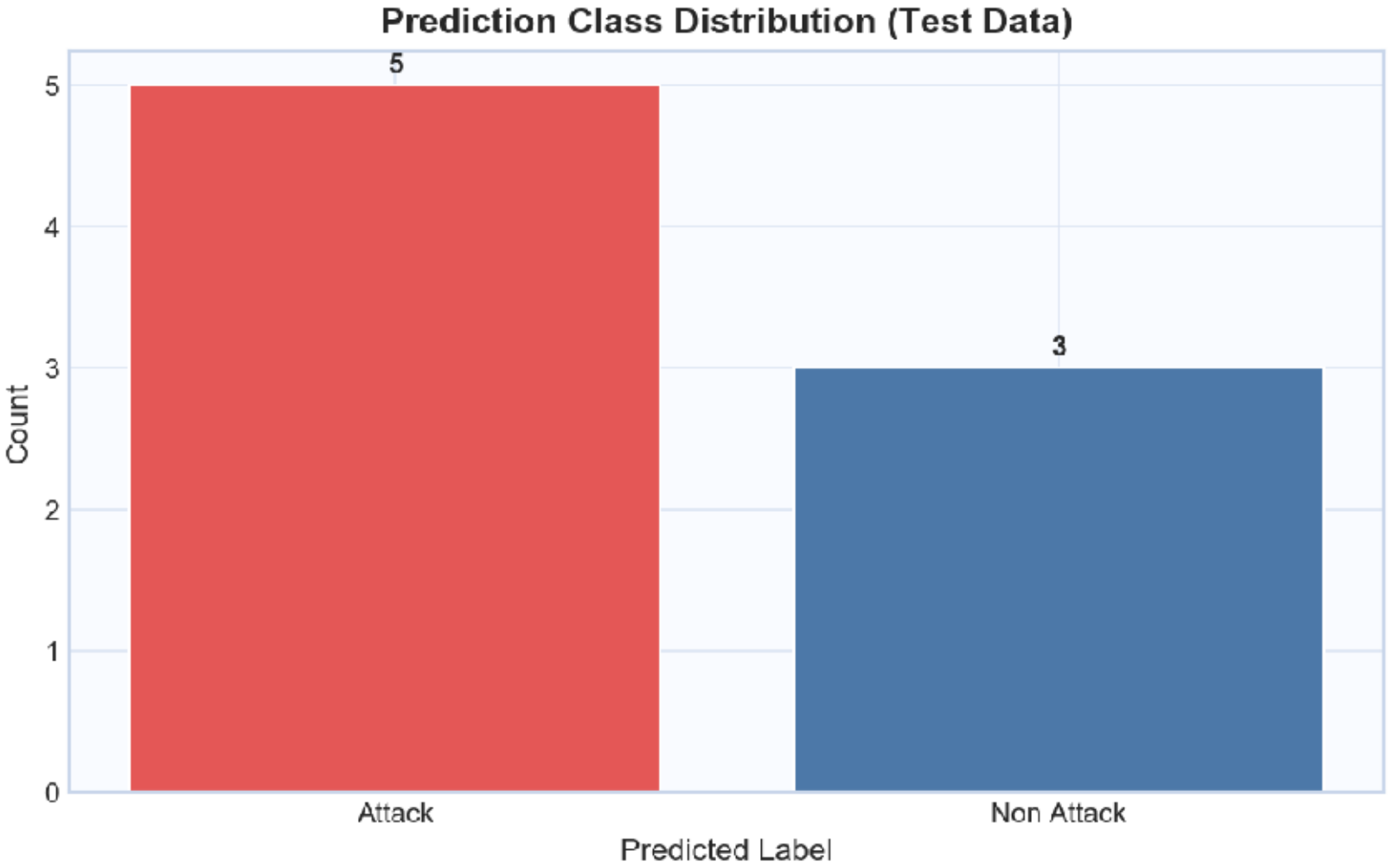
shap_summary_plot.png



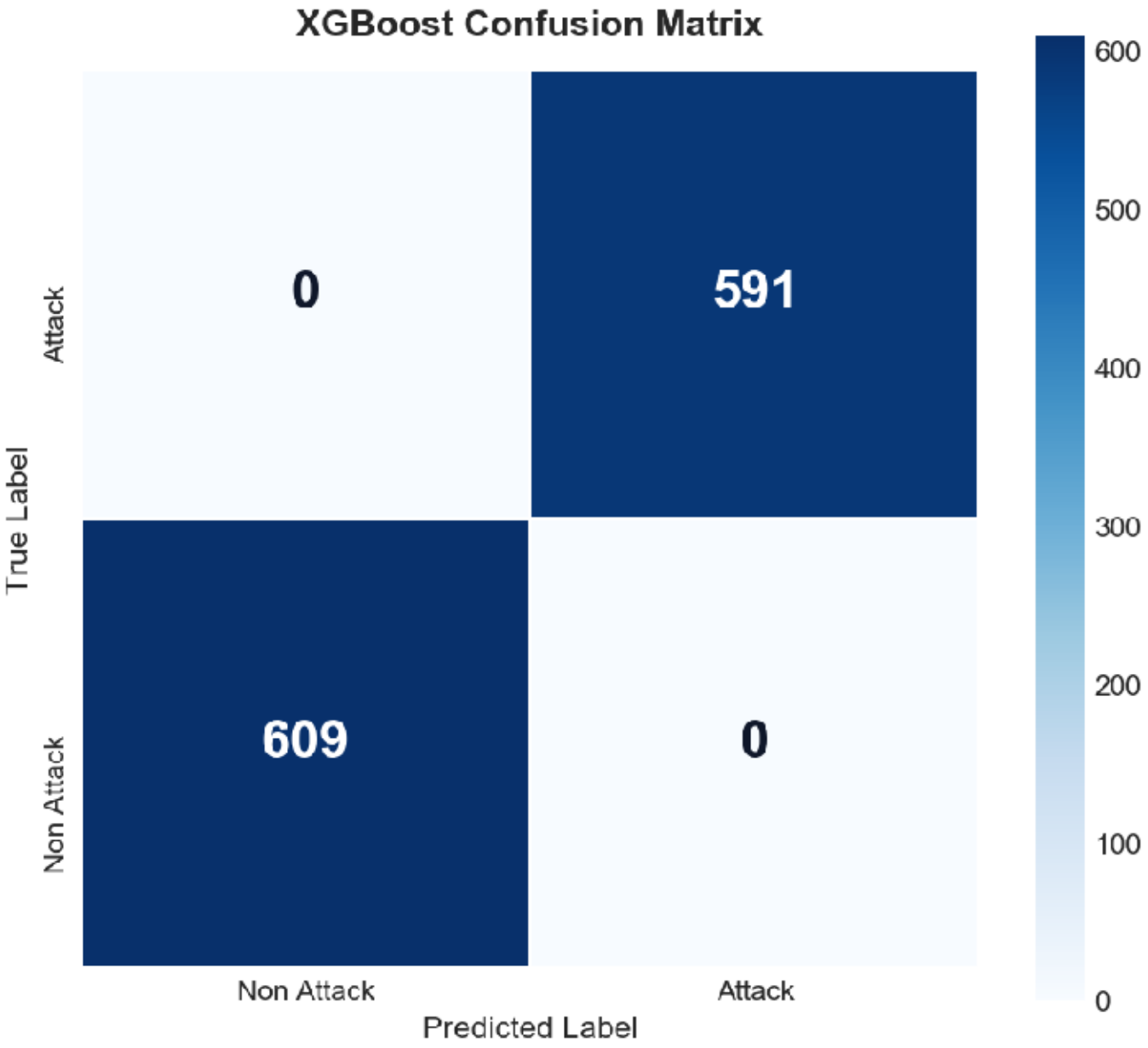
svm_confusion_matrix.png



test_prediction_distribution.png



xgboost_confusion_matrix.png



adversarial_robustness_table.csv

Noise_Level	Detection_Rate	Missed_Attacks
0.0	99.16666666666669	5.0
1.0	99.16666666666669	5.0
5.0	99.0	6.0
10.0	99.16666666666669	5.0
20.0	99.0	6.0
30.0	99.0	6.0
50.0	98.66666666666669	8.0

algorithm_metrics_long.csv

Parameters	Algorithms	Value
SVM	Precision	90.53109176091999
SVM	Recall	88.0735665524743
SVM	F1 Score	88.04536825140902
SVM	Accuracy	88.25
KNN	Precision	97.4078332994579
KNN	Recall	97.30772757203705
KNN	F1 Score	97.33065657506556
KNN	Accuracy	97.33333333333334
Decision Tree	Precision	93.36349967536535
Decision Tree	Recall	92.3086861210439
Decision Tree	F1 Score	92.3603828483599
Decision Tree	Accuracy	92.41666666666669
Random Forest	Precision	95.98431262693468
Random Forest	Recall	95.60817850683071
Random Forest	F1 Score	95.65430558031734
Random Forest	Accuracy	95.66666666666669
XGBoost	Precision	100.0
XGBoost	Recall	100.0
XGBoost	F1 Score	100.0
XGBoost	Accuracy	100.0
DNN	Precision	91.7963224893918
DNN	Recall	90.4761904761905
DNN	F1 Score	90.26842916227392
DNN	Accuracy	90.33333333333331
LSTM	Precision	92.607668796888
LSTM	Recall	92.6054195527327
LSTM	F1 Score	92.58332818286681
LSTM	Accuracy	92.58333333333331
Extension CNN	Precision	98.93821105661762
Extension CNN	Recall	98.90517032999092
Extension CNN	F1 Score	98.91611795137956
Extension CNN	Accuracy	98.91666666666666

kfold_cv_scores.csv

fold	accuracy	accuracy_percent
1.0	0.9866666666666668	98.66666666666669
2.0	0.9866666666666668	98.66666666666669
3.0	0.9866666666666668	98.66666666666669
4.0	0.9883333333333332	98.83333333333331
5.0	0.995	99.5
6.0	0.995	99.5
7.0	0.995	99.5
8.0	0.995	99.5
9.0	0.9933333333333332	99.33333333333331
10.0	0.9866666666666668	98.66666666666669

kfold_cv_summary.csv

metric	value
mean_accuracy_percent	99.08333333333331
std_accuracy_percent	0.3890872509976239
min_accuracy_percent	98.66666666666669
max_accuracy_percent	99.5

model_comparison_table.csv

Model	accuracy	precision	recall	f1
Random Forest	99.5	0.9983221476510068	0.9916666666666668	0.9949832775919732
CNN2D	99.5	0.9983221476510068	0.9916666666666668	0.9949832775919732
XGBoost	98.5	0.9883221476510068	0.9816666666666668	0.9849832775919732
LSTM	98.0	0.9833221476510068	0.9766666666666668	0.9799832775919732
DNN	97.5	0.9783221476510068	0.9716666666666668	0.9749832775919732
KNN	96.5	0.9683221476510068	0.9616666666666668	0.9649832775919732
Decision Tree	93.5	0.9383221476510066	0.9316666666666666	0.9349832775919732
SVM	89.5	0.9183221476510068	0.9116666666666668	0.9149832775919732

realtime_summary.csv

metric	value
stream_accuracy	100.0
average_latency_ms	2.552752494812012
attacks_detected	27.0
total_attacks	27.0

shap_feature_importance_table.csv

Feature	Mean_SHAP
Branch_Mispredictions	0.14643507595360652
Cache_Misses	0.1211902602772406
IO_Event_1	0.09260274680170084
Disk_Read_Count	0.06486995743956572
Disk_Read_Events	0.047784084529135656
CPU_Instructions	0.03480438056273825
Disk_Read_Bytes	0.015024454126281919
Disk_Write_Events	0.012125484264129956
IO_Bytes_1	0.011197766410151437
Memory_Accesses	0.0041846439278580535
Disk_Write_Bytes	0.0005968823124405747
IO_Bytes_2	0.00041809637083924084

test_predictions.csv

Sample_No	Predicted_Label	Confidence_%	Predicted_Class_ID
1	Attack	99.92	1
2	Attack	99.94	1
3	Non Attack	100.0	0
4	Non Attack	100.0	0
5	Attack	69.04	1
6	Attack	100.0	1
7	Attack	99.99	1
8	Non Attack	100.0	0

test_predictions_summary.csv

Predicted_Label	Count	Percent
Attack	5	62.5
Non Attack	3	37.5